

Assignment: 2

Name: Kaushal Kumar Ray

Branch: B Tech CSE with AI & ML

PRN: 240205241023

Subject: TOCE

Semester: IV

① Digital Innovation Environment

→ An ecosystem where digital technologies like AI, cloud, and IoT enable continuous development of new products, services, and business models.

② Transaction costs in Digital Ecosystems

→ costs involving in searching, negotiating, and executing transactions. Digital Platforms reduce these through automation and connectivity.

③ Role of Trust and Review systems.

→ They build user confidence by providing ratings, feedback, and transparency, helping users make informed decisions.

④ Personalisation in Digital Ecosystems.

- Assigning human-like traits to digital systems (e.g. chatbots, virtual assistants) to improve user interaction and engagement.

⑤ Artificial Intelligence Define it.

- Artificial intelligence is the ability of machines to simulate human intelligence, including learning, reasoning, and decision-making.

⑥ Explain the role of robotics in modern innovation systems.

- Robotics automate physical tasks, increases efficiency, and enhances precision in industrial like manufacturing and healthcare.

⑦ What is computing recognition?

- Computing recognition is the ability of systems to identify patterns such as speech, images, or text using AI techniques.

⑧ Explain the concept of decision-making systems.

→ Decision-making systems use data and algorithms to analyze situations and suggest or make decision automatically.

⑨ Define digital infrastructure in innovation ecosystems.

→ Digital infrastructure includes technologies like the internet, cloud computing, and data centers that support innovation.

⑩ What are cyberphysical systems?

→ Cyberphysical systems integrate physical processes with computational and digital control systems.

⑪ Explain the importance of AI in reducing transaction costs.

→ AI automates processes, reduce manual effort, and speeds up operations, lowering transaction costs.

⑫ Describe the role of digital technologies in open innovation.

→ Digital technologies enable collaboration, knowledge sharing, and innovation across organizations globally.

⑬ What are the components of cyberphysical systems?

→ They include sensors, actuators, computing units, communication networks, and control systems.

⑭ Explain the relationship between robotics and AI.

→ AI provides intelligence and decision-making while robotics executes tasks based on AI instructions.

⑮ Discuss the importance of digital infrastructure in innovation ecosystems.

→ It provides connectivity, scalability, and access to resources necessary for innovation and growth.

⑩ Explain the concept of digital technologies as an open innovation environment.

→ Digital technologies, allow organizations to collaborate externally and innovate beyond internal boundaries.

⑪ Discuss transaction costs and their role in digital economies.

→ Transaction costs are reduced through digital platforms, improving efficiency and enabling faster economic activities.

⑫ Explain the importance of trust and review systems in digital platforms.

→ They build credibility, reduce uncertainty, and encourage user participation in digital ecosystems.

⑬ Discuss the role of AI in modern innovation ecosystems.

→ AI drives automation, data analysis, personalization, and enhances innovation across sectors.

②⑥ Explain the application of robotics in industrial and business environments.

→ Robotics is used in manufacturing, logistics, healthcare, and services for automation and efficiency.

②⑦ Discuss computing recognition and decision-making systems.

→ Recognition systems identify patterns, while decision-making systems use that data to take or recommend actions.

②⑧ Explain the concept and structure of cyberphysical systems.

→ Cyberphysical system combine physical components with software and networks for real-time monitoring and control.

②⑨ Analyze how digital technologies reduce transaction costs in innovation ecosystems.

→ Digital tech reduce transaction costs by automating processes, enabling real-time communication, and minimizing intermediaries.

(24) Discuss the integration of robotics and AI in modern industries.

→ AI enables intelligent decision-making, while robotics performs tasks, together improving automation and productivity.

(25) Explain the importance of digital infrastructure in supporting innovation.

→ Digital infrastructure provides the necessary computing power, connectivity, and storage to support innovation.

(26) Discuss the role of cyberphysical systems in smart manufacturing.

→ They enable real-time monitoring, automation, and predictive maintenance, improving efficiency and reducing downtime.

(27) Explain how AI supports decision-making processes in business.

→ AI analyzes large datasets, predicts trends, and provides insights for faster and

accurate decision-making.

② Discuss the impact of digital transformation on innovation ecosystems.

→ Digital transformation enhances collaboration, enables new business models, and accelerates innovation.

③ Analyze the relationship between AI, robotics, and cyberphysical systems.

→ AI provides intelligence, robotics performs actions, and cyberphysical systems integrate both with physical environments.

④ Evaluate the role of digital technologies in building innovation infrastructure.

→ Digital technologies support connectivity, scalability and collaboration, forming the backbone of innovation infrastructure.

⑤ Rayan 19/3/26